

CATS – J OFFLINE – PRACTICE SHEET – 1

SECTION – 1 : (Maximum Marks : 28)

- This section contains SEVEN (07) questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct
- For each question, click the correct option(s) and save.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is(are) clicked and saved.
Partial Marks : +1 For clicking a bubble corresponding to **each correct option**, provided NO incorrect option is clicked.
Zero Marks : 0 If none of the bubbles is clicked.
Negative Marks : -2 In all other cases.
- For example, if (A), (C) and (D) are all the correct options for a question, clicking all these three will result in +4 marks; clicking only (A) and (D) will result in +2 marks and clicking (A) and (B) will result in -2 marks, as a wrong option is also clicked.

1. A satellite revolves around a planet in circular orbit of radius R (much larger than the radius of the planet) with a time period of revolution T . If the satellite is stopped and then released in its orbit (Assume that the satellite experiences gravitational force due to the planet only).

(A) It will fall into the planet

(B) The time of fall of the satellite is nearly $\frac{T}{\sqrt{8}}$

(C) The time of fall of the satellite into the planet is nearly $\frac{T\sqrt{2}}{8}$

(D) It cannot fall into the planet so time of fall of the satellite is meaningless

2. At the equator on a spherical planet a body weighs one third of that at the pole. Radius of planet is R .

The time period of revolution of the planet around its axis is equal to $T = 60$ min. (Take : $G = \frac{20}{3} \times 10^{-11} \text{ Nm}^2/\text{kg}^2$)

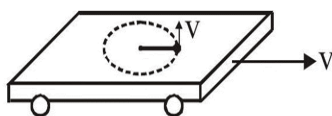
(A) The average density ρ of the planet is $\frac{\pi}{192} \times 10^6 \text{ kg/m}^3$

(B) The height of geostationary satellite from the surface of planet is $\left(\sqrt[3]{\frac{3}{2}} - 1\right) R$

(C) At a latitude of 60° , the weight of the body is half the weight at pole

(D) The geostationary satellite can be in a plane passing through the poles.

3. On a train moving along east with a constant speed v , a boy revolves a bob with string of length ℓ on smooth surface of a train, with equal constant speed v relative to train. Mark the correct option(s).



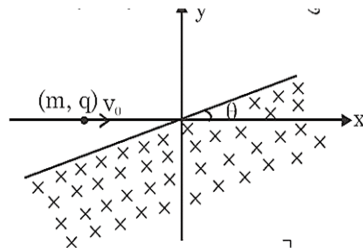
(A) Maximum speed of bob is $2v$ in ground frame.

(B) Tension in string connecting bob is $\frac{4mv^2}{\ell}$ at an instant.

(C) Tension in string is $\frac{mv^2}{\ell}$ at all the moments.

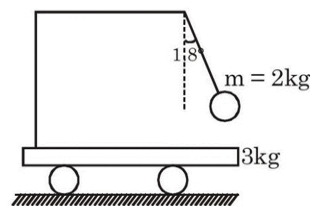
(D) Minimum speed of bob is zero in ground frame.

4. An ant travels along a long rod with a constant velocity $\vec{u}$ relative to the rod starting from the origin. The rod is kept initially along the positive x -axis. At $t = 0$, the rod also starts rotating with an angular velocity ω (anticlockwise) in $x - y$ plane about origin. Then
- (A) the position of the ant at any time t is $\vec{r} = ut[\cos \omega t \hat{i} + \sin \omega t \hat{j}]$
 (B) the speed of the ant at any time t is $u\sqrt{1 + \omega^2 t^2}$
 (C) the magnitude of the tangential acceleration of the ant any time t is $\frac{\omega^2 tu}{\sqrt{1 + \omega^2 t^2}}$
 (D) the speed of the ant at any time t is $\sqrt{1 + 2\omega^2 t^2}u$
5. A uniform magnetic field $-B_0 \hat{k}$ exists to the right of the plane $y = x \tan \theta$ as shown in the diagram. At $t = 0$, a particle of mass m and positive charge q with velocity $v_0 \hat{i}$ enters the magnetic field at origin. Mark the CORRECT statement (s) :-



- (A) Particle will come out from magnetic field after $t = \frac{\theta m}{qB_0}$
 (B) Particle will come out from magnetic field after $t = \frac{2\theta m}{qB_0}$
 (C) Co-ordinate of point from which particle will come out is $\left[\frac{mv_0}{qB_0} \sin 2\theta, \frac{m}{qB_0} (1 - \cos 2\theta), 0 \right]$
 (D) Co-ordinate of point from which particle will come out is $\left[\frac{mv_0}{qB_0} \sin \theta, \frac{mv_0}{qB_0} (1 - \cos \theta), 0 \right]$
6. There exist a uniform electric field $E\hat{j}$ and a uniform magnetic field $B\hat{k}$ and no gravitational field in region of space. A particle at origin of mass m , charge $+q$ is given velocity $\vec{v}_1 = +\frac{E\hat{i}}{B}$ or $\vec{v}_2 = \frac{+\sqrt{3}E\hat{j}}{B}$ at $t = 0$.
- (A) For particle velocity $\vec{v}_1 = +\frac{E\hat{i}}{B}$, electric force is balanced by magnetic force.
 (B) For particle velocity $\vec{v}_2$, particle will attain maximum y -coordinate for the first time at $t = \frac{2\pi m}{3qB}$
 (C) For particle velocity $\vec{v}_2$, maximum y -coordinate is $\frac{3mE}{qB^2}$.
 (D) For particle velocity $\vec{v}_2$, minimum y -coordinate is $\frac{-mE}{qB^2}$.
7. A simple pendulum of length $\ell = 30$ cm and mass $m = 2$ kg is attached to a cart of mass 3 kg. The cart can move frictionlessly on level ground. The bob is released from rest when string makes an angle of 1.8° with the horizontal :-

- (A) The amplitude of oscillation of cart is $\frac{\pi}{5}$ cm.
 (B) The time period of oscillation is $\frac{\pi}{5}$ sec
 (C) The cart and bob oscillate with phase difference of π .

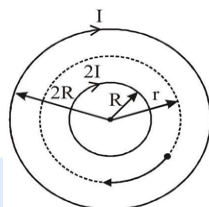


(D) The time period of oscillation is $\frac{\pi\sqrt{3}}{5}$ sec.

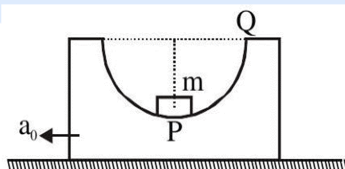
SECTION-2 (Maximum Marks: 09)

- This section contains **3** questions
- The answer to each question is a **DOUBLE DIGIT INTEGER** ranging from 00 to 99, both inclusive
- For each question, write the correct double digit integer and save/next
- Marking scheme :
 Full Marks : **+3** If the correct integer is saved
 Zero Marks : **0** If not attempted/Wrong Answer

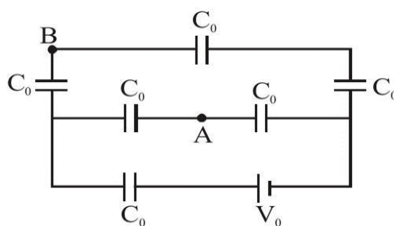
8. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have the same number of turns per unit length and initially both carry no current. At the same instant current starts increasing linearly with time in both solenoids. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing currents a charged particle, initially at rest between the solenoids, starts moving along a circular trajectory of radius r (see figure). The value of $\frac{r\sqrt{2}}{R}$ is.



9. A small block of mass m is lying at rest at point P of a wedge having a smooth semi circular track of radius R . The minimum value of horizontal acceleration a_0 (in m/s^2) of wedge so that mass can just reach the point Q , is



10. In the arrangement shown in figure, find the potential difference $V_B - V_A$ (in Volt). (Take $V_0 = 55 \text{ V}$)



SECTION 3 (Maximum Marks: 08)

- This section contains **TWO** questions.
- Each question has **TWO** matching lists: **LIST-I** and **LIST-II**.
- FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
- For each question, choose the option corresponding to the correct matching.
- For each question, marks will be awarded according to the following marking scheme:
 Full Marks : **+4** If **ONLY** the option corresponding to the correct matching is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

11. A charged particle with some initial velocity is projected in a region where uniform electric and/or magnetic fields are present. In Column-I information about the existence of electric and/or magnetic field and direction of initial velocity of charged particle are given, while in column-II the possible paths of charged particle is mentioned. Match the entries of Column I with the entries of Column-II.

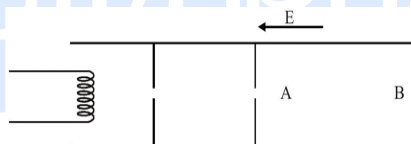
Column-I

- (A) $\vec{E} = 0, \vec{B} \neq 0$ and initial velocity is at an unknown angle with $\vec{B}$
(B) $\vec{E} \neq 0, \vec{B} = 0$ and initial velocity is at an unknown angle with $\vec{E}$
(C) $\vec{E} \neq 0, \vec{B} \neq 0, \vec{E} \parallel \vec{B}$ and initial velocity is perpendicular to $\vec{E}$
(D) $\vec{E} \neq 0, \vec{B} \neq 0, \vec{E}$ is perpendicular to $\vec{B}$ and initial velocity is perpendicular to both $\vec{E}$ and $\vec{B}$

Column-II

- (P) Straight line
(Q) Parabola
(R) Circular
(S) Helical path with non uniform pitch
(T) Helical path with uniform pitch

12. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of column-I with column-II as electron move from A to B :



Column-I

- (A) Speed of an electron
(B) Number of free electrons per unit volume
(C) Current density
(D) Electric potential

Column-II

- (P) Increases
(Q) Decreases
(R) Remains same
(S) any of the above is possible

CHEMISTRY

SECTION – 1 : (Maximum Marks : 28)

- This section contains **SEVEN (07)** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct
- For each question, click the correct option(s) and save.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is(are) clicked and saved.
Partial Marks : +1 For clicking a bubble corresponding to **each correct option**, provided NO incorrect option is clicked.
Zero Marks : 0 If none of the bubbles is clicked.
Negative Marks : -2 In all other cases.
- For example, if (A), (C) and (D) are all the correct options for a question, clicking all these three will result in +4 marks; clicking only (A) and (D) will result in +2 marks and clicking (A) and (B) will result in -2 marks, as a wrong option is also clicked.

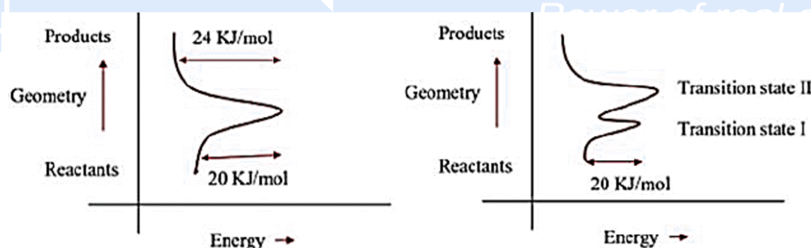
- Which of the following is/are correct for an ideal gas undergoing isothermal process? [All notations have their usual meaning]
(A) $\left(\frac{\partial G}{\partial P}\right)_T = +V$ (B) $\left(\frac{\partial A}{\partial V}\right)_T = -P$ (C) $\Delta H = 0$ (D) $w = q$
- Which of the following form(s) cyclic trimer or tetramer structures?
(A) SO_3 (B) SeO_3 (C) ClO_2 (D) HPO_3
- Which of the following give(s) white precipitate on addition to Na_2CO_3 solution?
(A) BaCl_2 (B) CaCl_2 (C) AgNO_3 (D) $\text{Cu}(\text{NO}_3)_2$
- When 5 mL of a 1.0M HCl solution is mixed with 5 mL of a 0.1M NaOH solution temperature of solution increases by $t^\circ\text{C}$. Which of the following can be predicted from this observation?
A) If 5 mL of 0.1M HCl is mixed with 5 mL 0.1M NH_3 solution, the temperature rise will be less than $t^\circ\text{C}$
B) If 5 mL 0.1M CH_3COOH is mixed with 5 mL 0.1M NaOH, the temperature rise will be less than $t^\circ\text{C}$
C) If 10 mL of 1.0M HCl is mixed with 10 mL of 0.1M NaOH, the temperature rise will be $2t^\circ\text{C}$
D) If 10 mL of 1.0M HCl is mixed with 10 mL of 0.1M NaCl, the temperature rise will be $2t^\circ\text{C}$
- Identify the correct relationship:
A) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex is square planar in geometry
B) $\text{Fe}(\text{CO})_x$ where x is generally 5
C) Zn^{2+} complexes with $\text{sp}^3 \text{d}^2$ hybridization are diamagnetic in nature

- D) Cr^{3+} complex are paramagnetic in nature
6. Choose the incorrect methods for each
- A) Ni : electrolysis process, Cu : van Arkel process, Zn : zone refining, Ga : zone refining
- B) Ni : mond's process, Cu : electrolysis, Zr : van Arkel process, Ga : zone refining
- C) Ni : electrolysis process, Cu : zone refining, Zr : van Arkel process, Ga : electrolysis
- D) Ni : electrolysis process, Cu : zone refining, Zr : van Arkel process, Ga : mond's process
7. Which of the following statement(s) is / are correct?
- A) KBr on heating with MnO_2 and concentrated H_2SO_4 liberates Br_2 and SO_2 gases
- B) NaBr on heating with concentrated H_2SO_4 liberates Br_2 and SO_2 gases
- C) KBr forms HBr with concentrated H_3PO_4 in open beaker
- D) KCl on heating with concentrated H_2SO_4 forms Cl_2 and SO_2 .

SECTION-2 (Maximum Marks: 09)

- This section contains **3** questions
- The answer to each question is a **DOUBLE DIGIT INTEGER** ranging from 00 to 99, both inclusive
- For each question, write the correct double digit integer and save/next
- Marking scheme :
 Full Marks : **+3** If the correct integer is saved
 Zero Marks : **0** If not attempted/Wrong Answer

8. For the reaction $R \rightarrow P$, energy change with geometry of reactants (R) to products (P) for uncatalysed (left) and catalysed (right) reactions are shown below:



If energy difference between two transition states in presence of catalyst is 10KJ/mol then find $-\ln(f)$ at 1000K where 'f' is fraction of molecules that could overcome energy barrier in the first step of products to reactants in catalysed reaction? Report your answer nearest integer.

$R = 8.3\text{J/molK}$.

9. An equimolar solid mixture of AlCl_3 , FeCl_3 and CrCl_3 is washed with a buffer solution of $\text{pH} = 8$. After precipitates are removed the maximum possible ratio of any two trivalent ion concentrations in the solutions is $a \times 10^b$. What is $(b - a)$?
- K_{sp} values : $\text{Al}(\text{OH})_3 = 2 \times 10^{-34}$, $\text{Fe}(\text{OH})_3 = 4 \times 10^{-39}$, $\text{Cr}(\text{OH})_3 = 2 \times 10^{-31}$
10. A 100ml solution (I) that contains KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$, is made slightly alkaline by adding Na_2CO_3 and excess H_2O_2 solution. The residue is filtered, and supernatant solution is distinctly yellow. This solution is heated to boil off hydrogen peroxide and then dilute sulphuric acid is added to make 200ml solution (II). If 20ml of this solution required, 300ml of 0.1N Mohr's salt solution, 10ml of solution (I)

required 550ml of 0.1N Mohr's salt solution to oxidize oxidant(s) in acidic medium. Find total number moles of MnO_4^- and $\text{Cr}_2\text{O}_7^{2-}$ ions in solution (I)?

11. Match the following type of rays given in Column I with their characteristics given in Column II. Select an appropriate answer from the codes given below.

Column I (Type of rays)	Column II (Characteristics)
A. α - ray	1. Negatively charged particles similar to electrons.
B. β - ray	2. X-ray, neutral in nature.
C. γ - ray	3. Two units of positive charge and four units of atomic mass.

Codes

- (a) $A \rightarrow 3; B \rightarrow 2; C \rightarrow 1$ (b) $A \rightarrow 1; B \rightarrow 2; C \rightarrow 3$
(c) $A \rightarrow 3; B \rightarrow 2; C \rightarrow 1$ (d) $A \rightarrow 3; B \rightarrow 1; C \rightarrow 2$

12. Match the Column I with Column II and select the correct option from the codes given below.

Column I (Compounds)	Column II (Oxidation number of nitrogen)
A. N_2O_3	1. -3
B. HNO_3	2. +5
C. NO	3. +3
D. NH_4OH	4. +2

Codes

- (a) $A \rightarrow 2; B \rightarrow 3; C \rightarrow 4; D \rightarrow 1$ (b) $A \rightarrow 1; B \rightarrow 2; C \rightarrow 3; D \rightarrow 4$
(c) $A \rightarrow 4; B \rightarrow 1; C \rightarrow 2; D \rightarrow 3$ (d) $A \rightarrow 3; B \rightarrow 2; C \rightarrow 4; D \rightarrow 1$

MATHEMATICS

SECTION – 1 : (Maximum Marks : 28)

- This section contains **SEVEN (07)** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct
- For each question, click the correct option(s) and save.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is(are) clicked and saved.
Partial Marks : +1 For clicking a bubble corresponding to **each correct option**, provided NO incorrect option is clicked.
Zero Marks : 0 If none of the bubbles is clicked.
Negative Marks : -2 In all other cases.
- For example, if (A), (C) and (D) are all the correct options for a question, clicking all these three will result in +4 marks; clicking only (A) and (D) will result in +2 marks and clicking (A) and (B) will result in -2 marks, as a wrong option is also clicked.

1. If $f : R \rightarrow [-1, 2], f(x) = \frac{x^2 + 9bx + 17}{ax^3 + x^2 + bx + 33}$ is onto function and $f'(d) = f'(e) = 0$, then which among the following option(s) is/are CORRECT ?

A) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{b+1} = 1$ can be $\frac{1}{2}$

B) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{b+1} = 1$ can be $\frac{\sqrt{5}}{2}$

C) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{a+1} = 1$ can be $\frac{\sqrt{3}}{2}$

D) eccentricity of curve $\frac{x^2}{(d+e)^2} + \frac{y^2}{a+1} = 1$ can be $\frac{\sqrt{7}}{2}$

2. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, then which among the following option(s) is/are CORRECT ?

A) $A^3 - A^2 = A - I$

B) $\text{Det}(A^{2010} - I) = 0$

C) $A^{50} = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$

D) $A^{50} = \begin{bmatrix} 1 & 1 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$

3. Let α, β are the roots of equation $x^2 - a_1x + 1 = 0$ and consider the sequence of numbers $a_r, (r \geq 0)$ with $a_0 = 1$ and $a_{r+1}^2 = 1 + a_r \cdot a_{r+2}$, then which among the following option(s) is/are CORRECT ?

A) $a_r + a_{r+2} = a_1 \cdot a_{r+1}$

B) $a_r + a_{r+2} = 2a_{r+1}$

C) $a_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}$

D) $a_n = \frac{\alpha^{n+1} + \beta^{n+1}}{\alpha + \beta}$

4. If $(1+x+x^2)^6 = A_0x^6 + (A_1x^2 + B_1)x^5 + (A_2x^4 + B_2)x^4 + \dots + (A_6x^{12} + B_6)$, then which among the following option(s) is/are CORRECT ?

A) $A_0 = 141$

B) $A_5 = 6$

C) $\sum_{i=1}^{i=6} (A_i + B_i) = 588$

D) $\sum_{i=1}^{i=6} (A_i + B_i) = 3^6$

5. Let $C_1 : x^2 + y^2 - 60x - 20y + k_1 = 0$ and $C_2 : x^2 + y^2 - 20x - 30y + k_2 = 0$. If ABC is a triangle with its vertices on C_1 . Let altitudes AD, BE and CF intersect C_1 again at points H, J and K respectively. If C_2 is in circle of $\triangle DEF$ and $D \equiv (8, 9)$, then which among the following option(s) is/are CORRECT ?

A) k_1 is equal to 375.

B) The distance of orthocenter of $\triangle ABC$ from vertex C must be $5\sqrt{2}$ units.

C) AD is equal to $7\sqrt{10}$ units.

D) Area of hexagon AKBHCJA is equal to 1050 sq. units.

6. Let a line parallel to the axis of the parabola $y^2 = 4x$ be drawn through point $P(\alpha - 5, \alpha)$ to meet the parabola at Q. Equation of tangent at Q is $x - 2y + 4 = 0$. Let T be a point on the tangent, and M, N be feet of perpendiculars on SQ and directrix respectively from point T, S is focus, then which among the following option(s) is/are CORRECT ?

A) $\alpha (\neq 0)$ is equal to -4

B) If $QM = 3$, then TN is 2

C) If two tangents are drawn from point P to the parabola touching it at A and B and circumcentre of $\triangle PAB$ is at (β, γ) , then γ is -1

D) If orthocenter of $\triangle PAB$ (as defined in option C) is at (a, b), then a + b equals to 3

7. An ellipse having eccentricity $\frac{3}{4}$ passes through the point $P(5, 4)$. If the nearer focus is (1, 1) and equation of tangent at P on the ellipse $4x + 3y - 32 = 0$, then

A) Length of latus rectum of ellipse is 17.5

B) Length of chord through (1, 1) parallel to the tangent at P is 17.5

C) Length of perpendicular from the focus other than (1, 1) on the tangent at P is 35

D) Locus of point of intersection of perpendicular tangents drawn to ellipse is

$$(x + 11)^2 + (y + 8)^2 = 575$$

SECTION-2 (Maximum Marks: 09)

- This section contains **3** questions
 - The answer to each question is a **DOUBLE DIGIT INTEGER** ranging from 00 to 99, both inclusive
 - For each question, write the correct double digit integer and save/next
 - Marking scheme :
Full Marks : **+3** If the correct integer is saved
Zero Marks : **0** If not attempted/Wrong Answer
-

8. P is any point and O being the origin. On the circle with OP as diameter two points Q and R are on same side of OP such that $\angle POQ = \angle QOR = \theta$. Let P, Q, R be z_1, z_2, z_3 such that

$$2\sqrt{3}z_2^2 = (2 + \sqrt{3})z_1z_3. \text{ The degree measure of } \theta \text{ is } \alpha^\circ, \text{ then } \frac{\alpha}{3} \text{ is (in degrees).}$$

9. If $\sin(\sin x + \cos x) = \cos(\cos x - \sin x)$, find the smallest positive possible value of $\frac{2\pi}{\sin x}$.

10. On a normal standard die one of the 21 dots from any one of the six faces is removed at random with each dot equally likely to be chosen. The die is then rolled. If the probability that the top face has an odd number of dots is $\frac{p}{q}$ where p and q are in their lowest form, find $\frac{(p+q)}{4}$.

SECTION 3 (Maximum Marks: 08)

- This section contains **TWO** questions.
- Each question has **TWO** matching lists: **LIST-I** and **LIST-II**.
- FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
- For each question, choose the option corresponding to the correct matching.
- For each question, marks will be awarded according to the following marking scheme:
 Full Marks : +4 If **ONLY** the option corresponding to the correct matching is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
 Negative Marks : -1 In all other cases.

11. Match the following

List- I		List- II	
(P)	The equation of tangent of the ellipse $\frac{x^2}{2} + y^2 = 1$ which cuts off equal lengths of intercepts on coordinate axis is $y = \pm x + a$, then a can be equal to	(1)	$-\sqrt{3}$
(Q)	The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex then m can be equal to	(2)	$-\sqrt{2}$
(R)	The equation of the common tangent to parabola $y^2 = 4x$ and $x^2 = 4y$ is $x + y + \frac{k}{\sqrt{3}} = 0$, then k is equal to	(3)	$\sqrt{2}$
(S)	An equation of common tangent to parabola $y^2 = 8x$ and the hyperbola $3x^2 - y^2 = 3$ is $2x + \frac{ky}{\sqrt{2}} + 1 = 0$, then k can be equal to	(4)	$\sqrt{3}$

- A) $P \rightarrow 1, 4; Q \rightarrow 2, 3; R \rightarrow 4; S \rightarrow 2, 3$ B) $P \rightarrow 1, 3; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1, 4$
 C) $P \rightarrow 4; Q \rightarrow 1, 3; R \rightarrow 2, 3; S \rightarrow 1$ D) $P \rightarrow 1, 4; Q \rightarrow 2, 3; R \rightarrow 1; S \rightarrow 3, 4$

12. Match the Following :

List- I		List- II	
(P)	The number of rational points on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is	(1)	Infinite
(Q)	The number of integral points on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is	(2)	4
(R)	The number of rational points on the ellipse $\frac{x^2}{3} + y^2 = 1$ is	(3)	2
(S)	The number of integral points on the ellipse $\frac{x^2}{3} + y^2 = 1$ is	(4)	0

- A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 3$ B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$
 C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 3$ D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

ANSWER KEY - PHYSICS

- | | | | |
|--|-----------------|--------------|--------------|
| 1. (A, C) | 2. (A,B) | 3. (A,C,D) | 4. (A, B, C) |
| 5. (B,C) | 6. (A, B, C, D) | 7. (A, B, C) | 8. 02 |
| 9. 10 | 10. 05 | | |
| 11. (A) $\rightarrow$ (P, R, T); (B) $\rightarrow$ (P, Q); (C) $\rightarrow$ (S) ; (D) $\rightarrow$ (P) | | | |
| 12. A-P; B-Q; C-R; D-P | | | |

ANSWER KEY - CHEMISTRY

- | | | | |
|----------|------------|---------|---------|
| 1. A,B,C | 2. A, B, D | 3. A, B | 4. AB |
| 5. ABCD | 6. ACD | 7. ABC | 8. (04) |
| 9. (02) | 10. (0.1) | 11. (D) | 12. (D) |

ANSWER KEY - MATHEMATICS

- | | | | |
|--------|--------|---------|--------|
| 1. ABC | 2. ABC | 3. AC | 4. ABC |
| 5. ACD | 6. BD | 7. ABCD | 8. 05 |
| 9. 08 | 10. 08 | 11. A | 12. B |